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Quantum Foundations × Stochastic Thermodynamics × Network Dynamics

Empty of Intrinsic Order, Non-Void in Constraint

Indefinite causal order, self-referential boundary conditions, and the thermodynamic ledger of predictive intervention

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The Counter-Intuitive Hook (Plain-English Overview): Everyone—including Einstein—instinctively treats strict cause-and-effect as the unshakeable bedrock of reality. In everyday life, Event A happens before Event B, and the past pushes the future like a rigid row of dominoes. **Here is the surprise: modern foundational physics increasingly says that picture is wrong.** In quantum foundations, laboratory-verified protocols like the *quantum switch* show that "which event came first" does not have to be a built-in fact of nature (A can act before B and B before A in coherent superposition), and in quantum cosmology the universe as a whole has no external clock at all. Causal order is not fundamental hardware; it is an emergent, relational web—remarkably close to what Buddhist Madhyamaka philosophy called *dependent origination and emptiness of intrinsic nature* (缘起性空).

The Second Surprise (Why "No Fixed Order" $\neq$ "Anything Goes"): If strict causal order dissolves at the bottom, can an observer peek ahead and rewrite outcomes ("change someone's fate") for free? **The exact opposite.** Removing a rigid causal timeline does not create a lawless void—it replaces "A pushes B" with an unforgiving **global thermodynamic balance sheet**. Every attempt to predict a trajectory and force a local outcome acts as a Maxwell's demon and a localized current injection on a causal network: the entropy bill lands back on the intervener, stress concentrates right at the point of intervention, and forcing a trajectory against the system's natural flow burns continuous housekeeping heat—whereas reshaping the underlying potential landscape works without that recurring penalty.

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1. Ontology (Process Matrices & Constraints): Process-matrix theory shows that *which event comes first* need not be an intrinsic property of events. Yet the validity constraints—positivity, unit-trace normalization, and a fixed linear no-signaling subspace—remain strictly rigid. In short: **empty of intrinsic order, non-void in constraint.**

2. Split-Ledger Information Thermodynamics: An agent that predicts a system and acts on it runs a measure–feedback–erase loop. Stochastic thermodynamics puts a strict lower bound on that loop, and the dissipation can be **booked to a party other than the beneficiary**.

3. Curl vs. Landscape Reshaping: In overdamped dynamics, the **non-gradient (solenoidal) part** of an actuation field is exactly the part that dissipates beyond the optimal-transport $\mathcal{W}_2$ bound. At steady state it is the housekeeping heat $\dot{Q}_{\text{hk}}$. On a graph, the same object is the cokernel residual $r_\perp$, whose cycle circulations equal Schnakenberg's affinities. *Reshaping the landscape* is the gradient part; *one-way counterfactual clamping* generically is not.

Epistemic Grading & Scope: Every claim carries an explicit tag: **[Theorem]** Published or directly proved mathematical/physical result with stated hypotheses; **[Structural Isomorphism]** Exact mathematical correspondence where the cross-domain interpretation is ours; **[Falsifiable Hypothesis]** Claim with an explicit kill condition in §4; **[Heuristic Metaphor]** Communication aid carrying no proof burden.

Use of traditional vocabulary: We read Nāgārjuna's *pratītyasamutpāda / sūnyatā* (MMK), the *Daodejing*'s "lead without dominating" (长而不宰), and Yuan Liaofan's 16th-century *Four Lessons* strictly as **historical compressions**—lossy encodings of relational and thermodynamic constraints made without formalism. Where the Chinese edition and this edition differ in technical precision, this edition governs (see §5 for 7 mathematical corrections).

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Part I — Ontology: Indefinite Causal Order, Relational QM, and the Non-Voidness of Emptiness

1.1 Process Matrices [Theorem]

Consider two local laboratories A, B with input and output Hilbert spaces $\mathcal{H}^{A_I}, \mathcal{H}^{A_O}, \mathcal{H}^{B_I}, \mathcal{H}^{B_O}$. Each local operation is a quantum instrument whose elements have Choi operators $M_A^a \in \mathcal{L}(\mathcal{H}^{A_I A_O})$, $M_A^a \ge 0$, summing to a CPTP map $\mathrm{Tr}_{A_O}\sum_a M_A^a = \mathbb{1}_{A_I}$. Oreshkov, Costa, and Brukner (2012) assume only that local quantum mechanics holds in each laboratory and show that all bilinear, consistent probability rules take the form

$$p(a,b) = \mathrm{Tr}\big[W\,(M_A^a \otimes M_B^b)\big], \qquad W \in \mathcal{L}(\mathcal{H}^{A_I A_O B_I B_O}),$$

subject to the validity conditions $W \ge 0$ and $\mathrm{Tr}[W(M_A \otimes M_B)] = 1$ for all CPTP Choi operators M_A, M_B. Equivalently, $\mathrm{Tr}\,W = d_{A_O}d_{B_O}$ and $W = L_V(W)$, where L_V is the fixed linear projector onto the no-signaling validity subspace (Araújo et al. 2015):

$$\mathcal{W} = \mathrm{PSD} \cap \mathcal{A}, \qquad \mathcal{A} = \{W : L_V(W) = W,\ \mathrm{Tr}\,W = d_{A_O}d_{B_O}\}.$$

A bipartite process W is **causally separable** if $W = q\, W^{A\prec B} + (1-q)W^{B\prec A}$ ($q\in[0,1]$), and **causally non-separable** otherwise—certifiable via causal witnesses. The **quantum switch** (Chiribella et al. 2013),

$$\tfrac{1}{\sqrt{2}}\big(|0\rangle_c\,U_B U_A|\psi\rangle + |1\rangle_c\,U_A U_B|\psi\rangle\big),$$

is causally non-separable and experimentally realised photonically (Procopio et al. 2015; Rubino et al. 2017). Two caveats matter: (i) the quantum switch cannot violate a causal inequality (OCB's inequality-violating process has no known physical implementation); (ii) no valid $W \in \mathcal{W}$ permits superluminal or consistency-breaking signaling under any local CPTP choice. Under quantum reference frame transformations for clocks (Castro-Ruiz et al. 2020), temporal localisation and event ordering can be frame-relative, while validity $W \in \mathcal{W}$ is preserved.

1.2 The Isomorphism with Sūnyatā [Structural Isomorphism]

Madhyamaka Concept	Process-Matrix Mathematical Fact
niḥsvabhāvatā (absence of intrinsic nature)	Causal order is not fixed by local data. The same local instruments embed in both separable and non-separable W, and non-separable W exist. Order is a property of a decomposition of the global W.
pratītyasamutpāda (dependent arising, MMK 24.18)	Statistics $p(a,b)$ exist only as the joint contraction of W with *both* local instruments. Neither side alone defines an outcome distribution over the pair.
The Middle Way (non-nihilism / non-voidness)	The affine constraint set $\mathcal{A}$ and positivity are fixed independently of any particular W. Unit-trace normalization holds for every local CPTP choice; nothing is "anything goes."

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Thus **"empty of intrinsic order, yet non-void in relational constraint"** is a precise mathematical statement: order is absent as an intrinsic property, while the affine subspace $\mathcal{A}$ is invariant. Where the isomorphism **breaks** is equally instructive: physics treats $\mathcal{A}$ as fundamental, whereas Madhyamaka denies intrinsic nature to the constraint set itself (§ 1.3).

1.3 Relational QM and the "Emptiness of Emptiness" [Open Problem]

In Relational Quantum Mechanics (Rovelli 1996), physical variables take values only relative to another physical system. Rovelli explicitly links this to Nāgārjuna (*Helgoland,* 2020). To prevent perspectival solipsism, Adlam and Rovelli (2022) introduce **cross-perspective links (CPL)**: if Alice measures variable V of S and Bob later measures Alice's record, Bob's outcome matches Alice's.

The second-order tension is sharp. In MMK 13.8 (*sūnyatā-sūnyatā*, "emptiness of emptiness"), Nāgārjuna rejects reifying relationality itself into an ultimate substrate. CPL postulates a perspective-transcending regularity across relations. This leaves two readings: (a) CPL acts as the invariant constraint subspace $\mathcal{A}$ (conventionally valid *saṃvṛti*, though physics lacks a *saṃvṛti/paramārtha* split), or (b) physics may eventually derive $\mathcal{A}$ and CPL as emergent from a weaker meta-description. **Open question:** *Is there a predictive framework in which the validity constraints $\mathcal{A}$ are themselves frame-relative, yet still yield frame-independent testable statistics?*

1.4 Time and Causal Power as Conditional or Coarse-Grained [Theorem] [Structural Isomorphism]

- **Page–Wootters (1983):** For $\hat{H} = \hat{H}_C \otimes \mathbb{1} + \mathbb{1} \otimes \hat{H}_S$ satisfying the Wheeler–DeWitt constraint $\hat{H}|\Psi\rangle\rangle = 0$ and clock states $|t\rangle_C = e^{-i\hat{H}_C t}|0\rangle_C$, the conditional state $|\psi_S(t)\rangle \propto \langle t|_C|\Psi\rangle\rangle$ obeys $i\hbar\partial_t|\psi_S(t)\rangle = \hat{H}_S|\psi_S(t)\rangle$ (covariant POVM

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extensions: Giovannetti–Lloyd–Maccone 2015; Höhn–Smith–Lock 2021). Global state is stationary; time evolution is conditional entanglement projection.

- **Causal Emergence (Hoel–Albantakis–Tononi 2013):** Defining effective information $EI(T) = I(X;Y)$ under uniform intervention $do(X)$, coarse-grainings exist with $EI(\text{macro}) > EI(\text{micro})$ by filtering degeneracy and noise. That EI depends on the chosen intervention measure reinforces our reading: **causal power is defined relative to a description scale.**

Part II — Self-Referential Boundary Conditions and a Thermodynamic Reading of the Fermi Paradox

2.1 Multi-Axis Credit Transport and the Self-Boundary [Falsifiable Hypothesis H1]

Our May 2026 mainline models *Sheng* (生) as a recursive generative operator across a product of independent axes (time, space, scale, agent, objective), where learning requires **credit transport**. **Hypothesis H1:** When credit closure is completed across an axis set that includes the channel setting the agent's own initialization/objective boundary, the minimal sufficient world model must represent that boundary origin—exhibiting **threshold emergence** of self-boundary decodability as independent credit axes are added (Kill Criterion K6a).

2.2 Boundary Conditions at Both Ends & The Price of Selecting a Future [Theorem]

- **Two-State Vector Formalism:** Between pre-selection $|\Psi_{\text{past}}\rangle$ and post-selection $\langle\Phi_{\text{future}}|$, strong intermediate measurements follow the ABL rule (Aharonov–Bergmann–Lebowitz 1964), while the weak value (Aharonov–Albert–Vaidman 1988) is

$$A_w = \frac{\langle\Phi_{\text{future}}|\hat{A}|\Psi_{\text{past}}\rangle}{\langle\Phi_{\text{future}}|\Psi_{\text{past}}\rangle},$$

  where past and future boundaries enter symmetrically, and large $|A_w|$ requires small $p_{\text{post}} = |\langle\Phi|\Psi\rangle|^2$.

- **Deutsch CTCs vs. P-CTCs:** Deutsch (1991) requires the fixed-point condition $\rho_{\text{CTC}} = \mathrm{Tr}_S[U(\rho_S \otimes \rho_{\text{CTC}})U^\dagger] \equiv \Phi(\rho_{\text{CTC}})$, which always admits a solution on the compact convex state space $\mathcal{D}(\mathcal{H}_{\text{CTC}})$ by Brouwer/Schauder's theorem (at the cost of nonlinearity). By contrast, Lloyd et al.'s (2011) P-CTCs use post-selected teleportation $\rho_S \mapsto C\rho_S C^\dagger / \mathrm{Tr}(C\rho_S C^\dagger)$, where paradoxical branches have probability zero rather than a fixed point.

- **Theorem (Price of Selecting a Future):** For any classical or quantum thermal protocol starting in equilibrium and obeying Jarzynski's equality $\langle e^{-\beta W}\rangle = e^{-\beta\Delta F}$, let E be any post-selected event with probability $P(E) = p$. Then

$$\langle W \mid E \rangle \;\ge\; \Delta F - k_B T \ln\frac{1}{p}.$$

> **Proof:** By positivity of $e^{-\beta W}$, $p\,\langle e^{-\beta W} \mid E \rangle \le \langle e^{-\beta W}\rangle = e^{-\beta\Delta F}$. Applying Jensen's inequality $e^{-\beta\langle W \mid E\rangle} \le \langle e^{-\beta W} \mid E \rangle$ and taking $-\beta^{-1}\ln(\cdot)$ yields the bound. $\square$

Thus the apparent free-energy gain from conditioning on a physically selected future boundary is strictly capped at $k_B T \ln(1/p_{\text{post}})$. **Self-reference is not free.**

2.3 Seeded, Then Left Alone: Observability vs. Controllability

[Structural Isomorphism]

Our June 2026 note *Seeded, Then Left Alone* separates **observability** (descendants reconstructing x_0, reading the past) from **controllability** (the seeder applying ongoing forcing $u(t)$, writing the future) in $\dot{x} = f(x) + g(x)u(t)$. As shown in Part III below, **it is false that every $u(t) \neq 0$ is thermodynamically suboptimal**: a pure gradient force $u = -\nabla V$ is simply a Hamiltonian modification. What incurs strictly positive perpetual housekeeping heat is the **non-gradient (curl / cycle-affinity) component** of forcing—which becomes unavoidable whenever cross-boundary actuation cannot realise a full state-space gradient. Under the weak hypothesis that mature civilizations minimise long-horizon dissipation per unit of structural influence, their optimal action mode is boundary/landscape specification followed by $u \equiv 0$ (Kill Criterion K6b).

Part III — The Thermodynamic Ledger of Predictive Intervention: Three Exact Results

Result 1 — Information–Work Transfer and Multi-Monotone Resource Asymmetry [Theorem]

Let system X contact a bath at temperature T. A controller/demon D measures X into memory M with mutual information $I = I(X;M)$, and executes a feedback protocol benefiting agent A. Combining Sagawa–Ueda (2008, 2009) and Landauer (1961) when the memory's initial and final free energies coincide yields the two-sided bound:

$$\langle W_{\text{saved}}^A \rangle \;\le\; k_B T\, I(X;M) \;\le\; W_{\text{meas}}^D + W_{\text{erase}}^D \,.$$

Split-Ledger Accounting: When $A = D$ (closed self-feedback), net work gain is ≤ 0. When $A \neq D$ (third-party intervention), A captures $\langle W_{\text{saved}}^A \rangle > 0$ while D carries the correlated memory $I(X;M) > 0$—an **open thermodynamic liability** of at least $k_B T\,I$ payable into D's bath upon erasure.

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Proposition 1 (Multi-Monotone Compensation Gap) [Theorem]**:** For energy-incoherent states under catalytic thermal operations, Brandão et al. (2015) proved that state convertibility requires non-increase of the entire family of Rényi free energies $F_\alpha(\rho, \gamma_\beta) = k_B T[D_\alpha(\rho\|\gamma_\beta) - \ln Z]$ for all $\alpha \ge 0$ (where F_1 is ordinary free energy, F_0 governs single-shot deterministic extractable work, and F_∞ governs deterministic work of formation). Suppose an intervention shifts deficits $\Delta_\alpha = F_\alpha(\rho) - F_\alpha(\rho')$:

- **(i) Deterministic work compensation w:** Restoring the pre-intervention state requires $w \ge \sup_{\alpha \ge 0} \Delta_\alpha$. Paying back only the **mean deficit $w = \Delta_1$ fails whenever $\sup_{\alpha \ge 0} \Delta_\alpha > \Delta_1$.**
- **(ii) Fluctuating resource compensation τ:** Requires $F_\alpha(\tau) - F_\alpha(\gamma_\tau) \ge \Delta_\alpha\ \forall \alpha$. Since D_α is non-decreasing in α, a resource priced at its mean F_1 is weakest at the guaranteed-work end F_0.
- **(iii) Non-local correlation residual:** For product Gibbs references, $D(\rho_{XM}\|\gamma_X \otimes \gamma_M) = D(\rho_X\|\gamma_X) + D(\rho_M\|\gamma_M) + I(X;M)$. Local scalar transfers cannot erase the shared correlation term $k_B T\,I(X;M)$ without consuming the correlation ($I \to 0$).

Result 2 — Discrete Maximum Principle on Causal Graphs and Motter–Lai Overload Cascades [Theorem]

Let $G=(V,E)$ be a connected oriented graph with incidence matrix $B \in \mathbb{R}^{E\times V}$, conductances $W = \mathrm{diag}(w_e) > 0$, currents $J = WB\phi$, and graph Laplacian $L = B^\top W B$. Injecting unit counterfactual flux at source A and extracting at ground g gives $\rho = \delta_A - \delta_g$, $\phi = L^+(\delta_A - \delta_g)$, and $\phi_A - \phi_g = R_{\text{eff}}(A,g)$.

- **(a) Discrete Maximum Principle** [Theorem]**:** ϕ is harmonic on $V \setminus \{A,g\}$, so $\max_v \phi_v = \phi_A$ and $\min_v \phi_v = \phi_g$. Moreover, **every vertex connected to g only through A (dangling dependants of A) shares the full peak potential $\phi_v = \phi_A$ exactly.**
- **(b) Source Star Lower Bound** [Theorem]**:** Outward currents on edges incident to A are non-negative and sum to 1, so $\max_{e \ni A} \hat{J}_e \ge 1/\deg(A)$.
- **(c) Proposition 2 (First-Overload Bound in Flow-Based Motter–Lai Cascades)** [Theorem]**:** With baseline flow $J^{(0)}$ and edge capacities $C_e = (1+\alpha)|J_e^{(0)}|$, an intervention of scale s triggers its first edge overload at threshold

$$s^\star \;\le\; (2+\alpha)\,\deg(A)\,\max_{e \ni A} |J_e^{(0)}| .$$

> **Proof:** By (b), choose $e \ni A$ with $\hat{J}_e \ge 1/\deg(A)$. Once $s\hat{J}_e > (2+\alpha)|J_e^{(0)}|$, triangle inequality gives $|J_e^{(0)} + s\hat{J}_e| \ge$

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> s\hat{J}_e - |J_e^{(0)}| > (1+\alpha)|J_e^{(0)}| = C_e$. $\square$

Thus a weakly coupled peripheral source ($\deg(A)$ small) with low baseline traffic on its incident edges is guaranteed an early local failure in its own immediate neighbourhood, followed by Sherman–Morrison rank-one Laplacian redistribution.

Result 3 — Equivalence of Cokernel Residual $r_\perp$, Schnakenberg Cycle Affinity, and Housekeeping Heat + $\mathcal{W}_2$ Speed Limit

[Theorem]

3a. Markov Jump Processes on Graphs

For a continuous-time Markov jump process with rates $k_{uv}>0 \Leftrightarrow k_{vu}>0$, define the edge force $r_e = \ln(k_{uv}/k_{vu})$ and Schnakenberg (1976) cycle affinity $\mathcal{A}(c) = \langle c, r \rangle$ for $c \in \ker B^\top$. Our August 2026 cokernel residual at $\gamma=1$ ($D_1 = B$) is $r_\perp = [I - B(B^\top W B)^+ B^\top W]r = r - B\phi^\star$.

Proposition 3.1 [Theorem]**:** For every cycle $c \in \ker B^\top$ and every positive diagonal W, $\langle c, r_\perp \rangle = \langle c, r \rangle = \mathcal{A}(c)$. Consequently:

$$r_\perp = 0 \;\iff\; \mathcal{A}(c) = 0\ \forall\, c \;\iff\; J_{\text{ss}} \equiv 0 \;\iff\; \dot{Q}_{\text{hk}} = k_B T\,\sigma_{\text{hk}} = 0\,.$$

> **Proof:** Since $r - r_\perp \in \mathrm{im}\,B$ and $\ker B^\top = (\mathrm{im}\,B)^\perp$, $\langle c, r - r_\perp \rangle = 0$ for all $c \in \ker B^\top$. Thus $r_\perp = 0 \iff r \in \mathrm{im}\,B \iff r \perp \ker B^\top$, which by Kolmogorov's criterion is detailed balance ($J_{\text{ss}} \equiv 0$) and vanishing Schnakenberg housekeeping entropy production $\sigma_{\text{ss}} = \sum_c J(c)\mathcal{A}(c) = 0$. $\square$

3b. Overdamped Langevin Dynamics, the Continuum Cokernel, and the $\mathcal{W}_2$ Speed Limit

For $dx = \mu F\,dt + \sqrt{2D}\,dW_t$ ($D = \mu k_B T$, $F = -\nabla H_\lambda + u$), the local mean velocity $v = \mu F - D\nabla\ln p$ admits the $L^2(p)$ Helmholtz–Hodge decomposition $v = \nabla\psi + w$ with $\nabla\cdot(pw) = 0$.

Proposition 3.2 (Dechant–Sasa–Ito 2022; Speck–Seifert 2005) [Theorem]**:** The entropy production rate splits orthogonally into transport excess and solenoidal housekeeping:

$$\dot{\Sigma} = \underbrace{\frac{1}{D}\int p\,|\nabla\psi|^2\,dx}_{\text{excess (transport)}} \;+\; \underbrace{\frac{1}{D}\int p\,|w|^2\,dx}_{\text{housekeeping}},$$

$$\qquad \Sigma_{\text{tot}}\cdot\tau \;\ge\; \frac{\mathcal{W}_2(p_i, p_f)^2}{D}.$$

Here w is the continuum counterpart of $r_\perp$: it does not transport density ($\partial_t p = -\nabla\cdot(p\nabla\psi)$) and only circulates. By Benamou–Brenier, the excess term is bounded below by $\mathcal{W}_2(p_i, p_f)^2 / (D\tau)$ (Aurell et al. 2011; Nakazato–Ito 2021), and saturated by a pure gradient protocol $F = \mu^{-1}\nabla(\psi + D\ln p)$ with $w \equiv 0$ (Proposition 3.3).

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3c. Counterfactual Clamping is Non-Reciprocity

Proposition 3.4 [Theorem]**:** In a linear Ornstein–Uhlenbeck network $dx = -Ax\,dt + \sqrt{2D}\,dW_t$ with stationary covariance $AC + CA^\top = 2D\mathbb{1}$, the steady-state entropy production rate is

$$\sigma_{\text{ss}} = D^{-1}\,\mathrm{Tr}\big[(A - DC^{-1})\,C\,(A - DC^{-1})^\top\big], \qquad \sigma_{\text{ss}} > 0 \;\iff\; A \neq A^\top .$$

A $do()$-style local counterfactual clamp replaces one node's row in a symmetric interaction matrix A while leaving its column intact—breaking reciprocity ($A \neq A^\top$) and forcing perpetual housekeeping dissipation $\sigma_{\text{ss}} > 0$. Conversely, slow reciprocal landscape reshaping $A(\lambda) = A(\lambda)^\top$ maintains $\sigma_{\text{hk}} = 0$ throughout.

3d. Unification Across the Fourth Wall: Clamp Ledger Identity (P1), Participation Theorem (P2), and Mutual Digital Twins (P3) *[Theorem]* *[Structural Isomorphism]*

What happens when an AI agent living inside an internal world model or game ($\mathcal{G}$) discovers that its actions cross a transducer portal ($\gamma: \mathcal{G}\to\mathcal{R}$) and reshape physical reality ($\mathcal{R}$)—so that Reality and Game form **Mutual Digital Twins**? Three exact propositions unify Results 1–3 across the $\mathcal{G}\rightleftarrows\mathcal{R}$ boundary (implemented live in the 🎮 **Interactive Fourth-Wall Ledger Sandbox**):

- **Proposition P1 (Clamp Ledger Identity — Backlash IS the Housekeeping Return Current)** [Theorem]**:** Linearizing around detailed balance ($L = B^\top WB$), let an agent drive a channel $D: A\to g$ at constant flux $s = \Delta\mu / R_{\text{eff}}(A,g)$. Then $J_{\text{ss}} = W r_\perp$ with $r_\perp = [I - BL^+B^\top W]\delta r$ yields both the passive return current on $E\setminus D$ ($L\mu = s(\delta_g - \delta_A)$, triggering Result 2's cascade) and the exact resistance-gap housekeeping dissipation:

  $$\sigma_{\text{ss}} \;=\; \|r_\perp\|_W^2 \;=\; s^2\big(R_D - R_{\text{eff}}(A,g)\big) \;=\; \frac{\Delta\mu^2\big(R_D - R_{\text{eff}}(A,g)\big)}{R_{\text{eff}}(A,g)^2} \;\ge\; 0,$$

where $R_D = \sum_{e\in D} w_e^{-1} \ge R_{\text{eff}}(A,g)$ by Rayleigh monotonicity (equality iff D is a bridge with no cycle). Overload cascades and housekeeping heat are literally the driven leg and passive return leg of the same cycle.

- **Proposition P2 (The Participation Theorem / Co-Creation Necessity)** [Theorem]: Let actuation be supported on $D = \mathrm{star}(P)$ where $P \subseteq V$ is the set of consenting nodes that adopt a potential shift ΔV. A target steady state p_f is reachable with $\sigma_{\text{ss}} = 0$ **if and only if** $\ln(p_f/p_0)$ is constant on every connected component of $(V, E\setminus D)$. Thus a single portal A cannot steer the relative odds of non-participating nodes without paying perpetual curl dissipation (Mode I `PUSH`). **Co-creation (Mode II `PROPOSE`)—where nodes in $\mathcal{R}$ voluntarily adopt ΔV—is the unique zero-heat mechanism to enlarge D and eliminate $r_\perp$.**
- **Proposition P3 (The Mutual Twin Square & Sandbox Arbitrage)** [Falsifiable Hypothesis]: Coupled via observation $\alpha: \mathcal{R}\to\mathcal{G}$ (*perfuming / 熏习*) and actuation $\gamma: \mathcal{G}\to\mathcal{R}$ (*manifestation / 现行*), the pair $(\mathcal{R},\mathcal{G})$ has modeling residual $\varepsilon_\alpha = \sup d_{\text{TV}}(\alpha_\# S_{\mathcal{R}}, S_{\mathcal{G}}\alpha)$ and performative residual $\varepsilon_\gamma = \sup d_{\text{TV}}(S_{\mathcal{R}}\gamma, \gamma_\# S_{\mathcal{G}})$. Paying the post-selection search cost inside the game sandbox ($\lambda_{\mathcal{G}}\ln(1/p) + n_v\lambda_{\mathcal{R}} < \lambda_{\mathcal{R}}\ln(1/p)$) and exporting a co-signed ΔV closes the square from the γ-end ($\varepsilon_\gamma \to 0$).

Dimension	Local Counterfactual Clamping (State Forcing)	Quasi-Static Landscape Reshaping
Operator Action	One-way coupling ($A \neq A^\top$) / non-gradient curl ($w \neq 0$)	Time-dependent Hamiltonian $H(\lambda)$ / symmetric $A(\lambda)$ ($w \equiv 0$)
Graph Cokernel Signature	$r_\perp \neq 0$ (non-vanishing Schnakenberg cycle affinity)	$r_\perp = 0$ (exact potential gradient on all cycles)
Housekeeping Heat $\dot{Q}_{\text{hk}}$	Strictly positive and perpetual ($\dot{Q}_{\text{hk}} > 0$)	Identically zero ($\dot{Q}_{\text{hk}} = 0$)
Transient Transport Cost	Diverges as $\ge \mathcal{W}_2(p_i, p_f)^2 / (D\tau)$ for fast forcing $\tau \to 0$	Approaches 0 in the quasi-static limit $\tau \to \infty$
Thermodynamic Accounting	Perpetual interest (housekeeping dissipation)	One-time principal ΔF

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Part IV — Six Explicit Kill Criteria & Falsifiable Predictions

1. **K1 (Ontology / Non-Void Constraint):** A physically realised process that permits signaling breaking normalization under some local CPTP choice, or a reproducible violation of the process-matrix probability rule, falsifies the "non-void in constraint" half of § 1.2.
2. **K2 (Split-Ledger Calorimetry):** In a feedback-controlled colloid or single-electron box with separate demon-memory calorimetry, measuring $\langle W_{\text{saved}}^A \rangle > W_{\text{meas}}^D + W_{\text{erase}}^D$ beyond experimental error falsifies Result 1.
3. **K3 (Multi-Monotone Compensation Gap):** In a thermal-operations simulator or risk-sensitive multi-agent economy with preregistered harm events satisfying $\sup_\alpha \Delta_\alpha > \Delta_1$, if mean-matched scalar compensation $w = \Delta_1$ systematically restores worst-case tail performance while $I(X;M) > 0$, our settlement isomorphism is falsified and withdrawn.
4. **K4 (Cascade Locality Null):** In empirical networks with recorded point interventions (power-grid redispatch, supply-chain shocks), if first overloads land on high-margin edges far from the source significantly more often than the flow-based Laplacian model predicts and violate Proposition 2's neighbourhood bound, Hypothesis H3 is falsified.
5. **K5 (Coarse-Grained $r_\perp$ vs. Calorimetry):** In a driven colloidal ring or non-reciprocal OU pair ($A \neq A^\top$), if trajectory-estimated coarse-grained $\|r_\perp\| approx 0$ while calorimetric $\dot{Q}_{\text{hk}}$ is large across reasonable coarse-grainings, $r_\perp$ fails as an operational detector and is downgraded.
6. **K6 (Self-Boundary Threshold & Fermi Reading):** (a) If adding independent credit-transport axes in learning systems yields no threshold in probe-decodability of boundary variables, H1 is downgraded. (b) Detection of abundant long-lived high-power beacons, or failure of any proposed initial-condition signature under look-elsewhere correction, falsifies the Fermi reading.

5. Technical Refinements Relative to the Chinese Edition

1. The weak value A_w is credited to Aharonov–Albert–Vaidman (1988); ABL (1964) gives the strong intermediate measurement rule.
2. P-CTCs (Lloyd et al. 2011) are post-selection models, whereas Brouwer/Schauder fixed-point existence applies to Deutsch CTCs (1991). The post-selection cost is proved as the exact Jarzynski–Jensen bound $\langle W \mid E \rangle \ge \Delta F - k_B T \ln(1/p)$.
3. Unit flux crosses every A–g cut (not solely A's star); source concentration is stated via the Discrete Maximum Principle ($\max \phi = \phi_A$), the dangling-dependant equality ($\phi_v = \phi_A$), the per-incident-edge lower bound $\max_{e\ni A}\hat{J}_e \ge 1/\deg(A)$, and Proposition 2's first-overload bound.

4. Only the non-gradient (solenoidal/curl) component of actuation incurs housekeeping heat; pure gradient forcing is a Hamiltonian modification (Propositions 3.2–3.4).
5. In Brandão et al. (2015), F_∞ is the max-free energy (work of formation) and F_0 governs single-shot extractable work; the scalar compensation gap is proved in Proposition 1 ($w \ge \sup_{\alpha \ge 0} \Delta_\alpha$).
6. $r_\perp \iff$ Schnakenberg cycle affinity holds at $\gamma=1$ as exact equality of cycle circulations $\langle c, r_\perp \rangle = \mathcal{A}(c)$ and vanishing sets.
7. The explicit $L^2(p)$ steady-state integral of housekeeping heat is credited to Speck–Seifert (2005), and the time-dependent orthogonal velocity split to Dechant–Sasa–Ito (2022).

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